Cheat sheet - 2nd DSP exam/UFPA

Chapter 3 – Systems

Convolution in discrete (DT) and continuous (CT) time:

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k] = \sum_{k=-\infty}^{\infty} h[k]x[n-k]. \qquad (1)$$

$$y(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau. \int_{-\infty}^{\infty} h(\tau)x(t-\tau)d\tau. \qquad (2)$$

Convolution properties:

- Commutative: $a{*}b = b{*}a$
- Associative: $a{*}(b{*}c) = (a{*}b){*}c$
- Distributive: $a{*}(b+c) = a{*}b + a{*}c$
- Duration: $N_1 + N_2 - 1$ samples
- Indices: if $a[n] = b[n]{*}c[n]$, the valid products to obtain $a[n_a]$ are $b[n_b]c[n_c]$ with $n_a = n_b + n_c$

$$y(t) = x_1(t){*}x_2(t) = \mathcal{F}^{-1}\{X_1(f)X_2(f)\} \qquad (3)$$

$$y[n] = x_1[n]{*}x_2[n] = \mathrm{DTFT}^{-1}\{X_1(e^{j\Omega})X_2(e^{j\Omega})\} \qquad (4)$$

Periodic or circular convolution (both signals of period T):

$$x_1(t)\circledast x_2(t) = \frac{1}{T}\int_{<T>} x_1(\tau)x_2(t-\tau)\mathrm{d}\tau \qquad (5)$$

In discrete-time frequency domain (implicit period 2π):

$$x_1[n]x_2[n] \Leftrightarrow X(e^{j\Omega})\circledast X_2(e^{j\Omega}) \qquad (6)$$

$$y[n] = x_1[n]\circledast x_2[n] = \mathrm{FFT}^{-1}\{X_1[k]X_2[k]\} \qquad (7)$$

Complex exponentials are eigenfunctions of LTI systems. In continuous-time (also valid for DT):

$$e^{s_0 t} \to \boxed{\mathcal{H}} \to H(s_0)e^{s_0 t}.$$

Tabela 1: Nomenclature of special frequencies in filtering.

Symbol	Description
ω_c or f_c	Cutoff frequency: gain falls by $1/\sqrt{2}$ (-3 dB)
ω_n	Natural frequency, for example, of a resonator
ω_0	Center frequency of a pole
f_p	Passband frequency
f_r	Stopband (or rejection) frequency
F_s	Sampling frequency (in Hz or samples per sec.)
$F_s/2$	Nyquist (or folding) frequency

Quality factor of poles in CT: $Q \stackrel{\text{def}}{=} \frac{f_n}{\mathrm{BW}}$ and DT: $Q = \frac{\omega_n}{2\alpha}$.
Second-order Butterworth filter:

$$H(s) = \frac{\omega_0^2}{s^2 + \sqrt{2}\,\omega_0 s + \omega_0^2}.$$

Group delay in CT (DT is similar):

$$\tau_g(\omega) = -\frac{d\Theta}{d\omega}. \qquad (8)$$

If the deviation in passband should be within $[1-D_p, 1]$ (linear scale) the ripple in dB to be within $[-R_p, 0]$ is:

$$R_p = -20\log_{10}(1-D_p).$$

For stopband:

$$R_s = -20\log_{10} D_s.$$

Filter frequency scaling of a prototype $H(s)$ with cutoff frequency 1 rad/s:

$$G(s) = H(s)|_{s=s/\omega_c} \qquad (9)$$

such that ω_c is the cutoff frequency of the new filter $G(s)$.
Simplified notations for A/DT$\{\cdot\}$ conversion:

$$x[n] = x(t)|_{t=nT_s} \qquad (10)$$

DT signal $x[n]$ with DTFT $X(e^{j\Omega})$ is converted into a sampled signal $x_s(t)$ with Fourier transform $X_s(\omega)$ via a DT/S conversion,

$$X_s(\omega) = X(e^{j\Omega})|_{\Omega=\omega T_s} = X(e^{j\omega T_s}). \qquad (11)$$

$$H_s(\omega) = H_z(e^{j\Omega})|_{\Omega=\omega T_s} = H_z(e^{j\omega T_s}), \qquad (12)$$

Perfect reconstruction of $x[n]$ when $h(t)$ is a sinc:

$$x(t) = \mathrm{DT/S}\{x[n]\}{*}h(t) = \sum_{n=-\infty}^{\infty} x[n]\mathrm{sinc}\left(\frac{t}{T_s} - n\right). \qquad (13)$$

Power relation:

$$\mathcal{P}_s \stackrel{\text{def}}{=} \frac{\mathcal{P}_d}{T_s}. \qquad (14)$$

IIR design methods based on $H(s)$: a) Matched Z-transform or pole-zero matching: $z_i = e^{s_i T_s}$. b) Impulse invariance

$$H(z) = T_s\mathcal{Z}\{h(t)|_{t=nT_s}\} \qquad (15)$$

c) Step invariance:

$$\mathcal{Z}^{-1}\left\{H(z)\frac{1}{1-z^{-1}}\right\} = \mathcal{L}^{-1}\left\{H(s)\frac{1}{s}\right\}\Bigg|_{t=nT_s} \qquad (16)$$

d) Backward difference: $s \leftarrow (1-z^{-1})/T_s$
e) Forward difference: $s \leftarrow (1-z^{-1})/(T_s z^{-1})$
f) Bilinear (or Tustin's): $s \leftarrow 2F_s\frac{z-1}{z+1}$
Bilinear frequency warping

$$\omega = \frac{2}{T_s}\tan\left(\frac{\Omega}{2}\right) = 2F_s\tan\left(\frac{\Omega}{2}\right), \qquad (17)$$

$$\omega_a = \frac{2}{T_s}\tan\left(\frac{T_s\omega_d}{2}\right). \qquad (18)$$

To match a frequency using the bilinear transform:

$$F_s' = \frac{\omega_m}{2\tan\left(\frac{\omega_m}{2F_s}\right)}, \qquad (19)$$

In FIR design via windowing, ideal lowpass filter impulse response:

$$h_{\mathrm{LP}}[n] = \frac{\sin(\Omega_c n)}{\pi n} \quad \text{, with } h_{\mathrm{LP}}[0] = \frac{\Omega_c}{\pi} \qquad (20)$$

Impulse response for the ideal highpass filter:

$$h_{\mathrm{HP}}[n] = \delta[n] - h_{\mathrm{LP}}[n] \qquad (21)$$

Symmetric FIR has linear phase and can be decomposed as

$$H(e^{j\Omega}) = e^{-j\Omega\tau_g}A(\Omega)$$

Realizations of digital filters: direct form I, direct form II, transposed direct form II, SOS in cascade, SOS in parallel.

Chapter 4 – Spectral Analysis

Windowing $x_w[n] = x[n]w[n]$ corresponds to *periodic convolution* in frequency-domain:

$$X_w(e^{j\Omega}) = X(e^{j\Omega}) \circledast W(e^{j\Omega}), \tag{22}$$

Hamming window:

$$w[n] = 0.54 - 0.46\cos\left(\frac{2\pi n}{N-1}\right), \quad n = 0, \ldots, N-1. \tag{23}$$

"Periodic" *Hann* window

$$w[n] = 0.5 - 0.5\cos\left(\frac{2\pi n}{N-1}\right), \quad n = 0, \ldots, N-1. \tag{24}$$

Important performance metrics of windows: a) width of its main lobe, b) amplitude of its sidelobes, c) rate of sidelobe decay as frequency increases, d) and scalloping loss.

Expressions to investigate bin and non-bin-centered sinusoids, where $\alpha \in \mathbb{R}$ and $\Delta\Omega = 2\pi/N_{\text{fft}}$:

$$x[n] = A\cos\left(\alpha \frac{2\pi}{N_{\text{fft}}} n\right) \quad \text{and} \quad x[n] = A\cos\left(\alpha \Delta\Omega n\right). \tag{25}$$

DTFT $X_w(e^{j\Omega})$ of the windowed cosine $x_w[n] = x[n]w[n]$:

$$X_w(e^{j\Omega}) = \frac{A}{2}\left[\frac{\sin\left(\frac{N(\Omega+\Omega_c)}{2}\right)}{\sin\left(\frac{\Omega+\Omega_c}{2}\right)} e^{-j\frac{(\Omega+\Omega_c)(N-1)}{2}} + \frac{\sin\left(\frac{N(\Omega-\Omega_c)}{2}\right)}{\sin\left(\frac{\Omega-\Omega_c}{2}\right)} e^{-j\frac{(\Omega-\Omega_c)(N-1)}{2}}\right]. \tag{26}$$

Leakage is the redistribution of spectral energy caused by windowing a signal.

Picket-fence effect is the phenomenon whereby the DFT/FFT observes the continuous DTFT only at the FFT bin centers.

Scalloping loss is the reduction in the measured amplitude of a spectral component when its frequency lies between two adjacent DFT/FFT bin centers rather than at a bin center.

Improving FFT resolution $\Delta f = F_s/N_{\text{fft}}$ by increasing N_{fft} *cannot recover the leakage that occurred due to windowing.*

Tabela 2: Energy spectral density (ESD) functions. The units of $\mathcal{G}(f)$, $\mathcal{G}(\omega)/(2\pi)$ and $\mathcal{G}(e^{j\Omega})/(2\pi)$ are J/Hz, J/(rad/s) and J/rad, respectively.

Time	ESD definition	Property
Cont.	$\mathcal{G}(f) = \lvert X(f)\rvert^2$	$E = \int_{-\infty}^{\infty} \mathcal{G}(f)\mathrm{d}f$
Cont.	$\mathcal{G}(\omega) = \lvert X(\omega)\rvert^2$	$E = \frac{1}{2\pi}\int_{-\infty}^{\infty} \mathcal{G}(\omega)\mathrm{d}\omega$
Disc.	$\mathcal{G}(e^{j\Omega}) = \lvert X(e^{j\Omega})\rvert^2$	$E = \frac{1}{2\pi}\int_{<2\pi>} \mathcal{G}(e^{j\Omega})\mathrm{d}\Omega$

Tabela 3: Power spectral density (PSD) functions. The units of $S(f)$, $S(\omega)/(2\pi)$ and $S(e^{j\Omega})/(2\pi)$ are W/Hz, W/(rad/s) and W/rad, respectively.

Time	Definition	Main property
Cont.	$S(f) = \mathcal{F}_f\{R(\tau)\}$	$\mathcal{P} = \int_{-\infty}^{\infty} S(f)\mathrm{d}f$
Cont.	$S(\omega) = \mathcal{F}_\omega\{R(\tau)\}$	$\mathcal{P} = \frac{1}{2\pi}\int_{-\infty}^{\infty} S(\omega)\mathrm{d}\omega$
Discr.	$S(e^{j\Omega}) = \mathcal{F}\{R[\ell]\}$	$\mathcal{P} = \frac{1}{2\pi}\int_{<2\pi>} S(e^{j\Omega})\mathrm{d}\Omega$

PSD of continuous-time periodic deterministic signal:

$$S(f) = \sum_{k=-\infty}^{\infty} |c_k|^2 \delta(f - kF_0), \tag{27}$$

with Fourier Series coefficients c_k and period $T_0 = 1/F_0$.
PSD of discrete-time sinusoid $x[n] = A\cos(\Omega_1 n)$:

$$S(e^{j\Omega}) = \sum_{p=-\infty}^{\infty} \frac{\pi A^2}{2}\left[\delta(\Omega + \Omega_1 + p2\pi) + \delta(\Omega - \Omega_1 + p2\pi)\right]$$

For a WSS random process $\mathcal{X}(t)$ with PSD $S_x(f)$, the PSD of $\mathcal{Y}(t) = \mathcal{X}(t)e^{j2\pi f_c t}$ is $S_y(f) = S_x(f - f_c)$.
Mean-square spectrum (MS spectrum):

$$S_{\text{ms}}[k] = |\text{DTFS}\{x[n]\}|^2. \tag{28}$$

If a WSS process $\mathcal{X}(t)$ is the input of a LTI system $\mathcal{X}(t) \rightarrow \boxed{h(t)} \rightarrow \mathcal{Y}(t)$, then $S_y(f) = |H(f)|^2 S_x(f)$. In discrete-time: $S_y(e^{j\Omega}) = |H(e^{j\Omega})|^2 S_x(e^{j\Omega})$.
The *bilateral* PSD of a white noise is

$$S(f) = \mathcal{F}\left\{\frac{\mathcal{N}_0}{2}\delta(\tau)\right\} = \frac{\mathcal{N}_0}{2}, \quad \forall f \tag{29}$$

and its average power over a bandwidth BW is $\mathcal{P} = \mathcal{N}_0\text{BW}$.
For discrete-time white noise: $S_y(e^{j\Omega}) = \sigma_x^2 |H(e^{j\Omega})|^2$.
Periodogram is an approximation of $S(f)$ given by

$$\hat{S}[k] \stackrel{\text{def}}{=} \frac{1}{N\ \text{BW}} |\text{FFT}\{x_N[n]\}|^2. \tag{30}$$

$$\mathcal{P} \approx \Delta f \sum_{k=0}^{N-1} \hat{S}[k], \tag{31}$$

Relation between MS spectrum and PSD:

$$\hat{S}_{\text{ms}}[k] = \mathcal{P}_k = \Delta f \hat{S}[k]. \tag{32}$$

$$\hat{S}(e^{j\Omega}) = \sum_{k=-N+1}^{N-1} \hat{R}[k] e^{-j\Omega n}. \tag{33}$$

Learn to use Welch's method in Matlab:

```
[Sx,w]=pwelch(x,window,Num_overlap,N_fft,Fs,
   'twosided')
```

and Python:

```
from scipy.signal import welch
w,Sx=welch(x,window=window,nperseg=len(window),
        noverlap=Num_overlap,nfft=N_fft,
window=window,return_onesided=False)
```

Auto-regressive modeling: calculate *prediction-error* FIR filter $A(z) = 1 + \sum_{i=1}^{P} a_i z^{-i} = 1 - \tilde{A}(z)$, where $\tilde{A}(z) = -\sum_{i=1}^{P} a_i z^{-i}$ is the *predictor* filter that estimates

$$\tilde{y}[n] = -\sum_{i=1}^{P} a_i y[n-i], \tag{34}$$

such that $A(z)$ minimizes the power of the *prediction error* $x[n] = y[n] - \tilde{y}[n]$. Having $A(z)$, the estimated PSD is then:

$$\hat{S}(e^{j\Omega}) = |H(e^{j\Omega})|^2 = \frac{g^2}{|A(e^{j\Omega})|^2}. \tag{35}$$

Short-time Fourier transform (STFT) and spectrogram:

$$X(\tau, f) = \int_{-\infty}^{\infty} x(t)w(t-\tau)e^{-j2\pi f t}\mathrm{d}t \tag{36}$$